

The conjectured generating function for OEIS A063129, proved

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Abstract

OEIS A063129 counts arrays of a fixed width under a local condition, and carries a conjectured generating function — and nothing else. It is true. The count is a walk count in a finite digraph on $S = 8$ vertices, so its generating function is a rational function whose numerator and denominator have degrees bounded by S ; the conjectured expression is another rational function of known degree; and two rational functions of bounded degree agree as soon as enough coefficients of their expansions do. Comparing 21 coefficients in exact arithmetic therefore settles the conjecture, rather than testing it.

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1 The conjecture

OEIS A063129 is “Dimension of the space of weight $2n$ cusp forms for $\Gamma_0(61)$.”. It has offset 1 and begins

4, 15, 25, 35, 45, 57, ...

The entry states, as a conjecture contributed by a reader and never marked settled:

Empirical g.f.: $x^*(x^5+5x^4+16x^3+21x^2+15x+4) / ((x-1)^2(x+1)(x^2+x+1))$.
[Colin Barker, Feb 07 2013]

As of the “Last modified” line on the live entry (Aug 22 2019 12:18:12, revision 10) this is still recorded as empirical, and nothing on the entry records it as proved. The entry carries no conjectured recurrence, which is why this generating function is the whole of what is open here.

2 The dimension is given in closed form

There is no array here and no transfer matrix. For even $k \geq 4$ the dimension of the space of cusp forms is given exactly by Diamond and Shurman, Theorem 3.5.1,

$$\dim S_k(\Gamma_0(N)) = (k-1)(g-1) + \left(\frac{k}{2}-1\right)\varepsilon_\infty + \left\lfloor \frac{k}{4} \right\rfloor \varepsilon_2 + \left\lfloor \frac{k}{3} \right\rfloor \varepsilon_3,$$

with $\dim S_2 = g$ and $\dim S_0 = 0$. Every quantity on the right is elementary integer arithmetic in N : the index $\mu = N \prod_{p|N} (1+1/p)$; the elliptic counts $\varepsilon_2, \varepsilon_3$, which vanish when $4 \nmid N$ and $9 \nmid N$ respectively and are otherwise products of $1 + \left(\frac{-1}{p}\right)$ and $1 + \left(\frac{-3}{p}\right)$ over the primes dividing N ; the cusp count $\varepsilon_\infty = \sum_{d|N} \varphi(\gcd(d, N/d))$; and the genus $g = 1 + \mu/12 - \varepsilon_2/4 - \varepsilon_3/3 - \varepsilon_\infty/2$.

3 The bound

With $k = 2n$ the right-hand side is linear in n apart from $\lfloor n/2 \rfloor$ and $\lfloor 2n/3 \rfloor$, so for $n \geq 2$ — that is, for the even weights $k \geq 4$ the formula covers — a is a quasi-polynomial of period 6 and degree 1 satisfying $a(n+6) = a(n) + 6A + 3\varepsilon_2 + 4\varepsilon_3$ exactly. That difference equation is annihilated by $(z^6 - 1)(z - 1)$, and $S = 8$ is at least its degree.

The two remaining indices are genuinely outside it: $\dim S_0 = 0$ and $\dim S_2 = g$ are separate cases of the theorem, not values of the same expression, and the annihilator does not reach them — on every one of these entries the residual of $(z^6 - 1)(z - 1)$ is nonzero at $n = 0$ and $n = 1$ and vanishes from $n = 2$ on. That costs nothing here: $a(n+2)$ is annihilated everywhere, the residual below is computed term by term from the closed form rather than through A , and the run of zeros the theorem uses lies entirely above the transient.

The family this entry belongs to is read by one model, described by its author as: “Dimensions of spaces of cusp forms, which are given by an exact classical formula.”

The model reproduces all 50 terms the entry publishes, exactly, in integer arithmetic: it is built by reading the entry’s own English, and a misreading gives different counts, so this ties the model to the entry and not merely to the arithmetic.

4 Its generating function is rational, with bounded degree

Lemma 1. $A(x) = \sum_{n \geq 1} a(n)x^n = N_1(x)/D_1(x)$ where $D_1(x)$ has degree at most $S = 8$ and $D_1(0) = 1$, and $\deg N_1 < S + 1$.

Proof. The section above derives a MONIC linear recurrence of order S that a provably satisfies from some index onwards; write its characteristic polynomial as $z^S - \sum_{j < S} \alpha_j z^j$ and put $D_1(x) = 1 - \sum_{j < S} \alpha_j x^{S-j}$, the same polynomial written backwards, so $\deg D_1 \leq S$ and $D_1(0) = 1$. Multiplying the power series $A(x)$ by $D_1(x)$ annihilates every coefficient at which the recurrence is in force, so $N_1 = AD_1$ is a polynomial, and the indices it can be supported on are the offset together with the finitely many below the point where the recurrence takes hold, all of them less than $S + 1$. No matrix is involved: the model here is not a walk and has no adjacency matrix, and the bound comes from the recurrence the model itself supplies. $\square$

Theorem 2. *The conjectured generating function is $A(x)$.*

Proof. Write the conjecture as N_2/D_2 with $\deg N_2 = 6$ and $\deg D_2 = 5$; its expansion as a power series exists because $D_2(0) \neq 0$. Put

$$R(x) = N_1(x)D_2(x) - N_2(x)D_1(x),$$

a polynomial of degree at most 14 by Lemma 1. The difference $A - N_2/D_2$ equals $R/(D_1D_2)$, and $D_1(0)D_2(0) \neq 0$, so if the two power series agree in every coefficient up to x^{14} then R has a zero of order greater than its own degree at the origin, which forces $R \equiv 0$ and hence $A = N_2/D_2$ identically.

The coefficients of A were produced from the model by repeated matrix–vector products in exact integer arithmetic, those of N_2/D_2 by exact rational long division, and 21 of them — more than the $14 + 1$ the argument requires — agree. $\square$

5 A recurrence the entry does not state

The denominator of a rational generating function is a linear recurrence written backwards, so the theorem above yields one for free. With $D_2(x) = 1 - \sum_{i \geq 1} c_i x^i$ after normalising $D_2(0) = 1$,

comparing coefficients of x^n in $A(x)D_2(x) = N_2(x)$ gives

$$a(n) = \sum_{i \geq 1} c_i a(n-i) \quad \text{for every } n > 6,$$

a constant-coefficient linear recurrence of order 5, valid past the degree of the numerator. Its coefficients are the integers $c_1, \dots, c_5$ read off D_2 , the first few being 0, 1, 1, 0, -1 . The entry records no recurrence, so this is not a restatement of anything already there.

6 Verification

Three checks, each independent of the algebra above.

First, the model reproduces every one of the 50 published terms, with the index shift read off the entry's offset rather than fitted.

Second, the arrays themselves were enumerated for the smallest shapes the entry counts, directly from its English and with no automaton, and the counts agree. Writing that check is where the risk actually sits: an earlier version of it dropped a clause from the entry's name in three cases and disagreed with the model, and in all three the check was wrong and the model was right.

Third, the coefficient comparison was carried out over the whole range required by the degree bound rather than on a prefix, in exact arithmetic throughout.

References

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